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Synergistic fermentation of *Cordyceps militaris* and herbal substrates boosts grower pig antioxidant and immune function

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Abstract

Background Pathogenic infections can significantly impact the health of livestock. Traditionally, antibiotic growth promoters (AGPs) have been used in feed to enhance growth performance and disease control. However, concerns regarding antibiotic resistance have led to the exploration of traditional herbal medicine as a natural alternative, guided by the principle of medicine-food homology. The Taguchi method was employed to optimize the culture formula for cordycepin production, an active component of *Cordyceps militaris* (*C. militaris*). The influences of *C. militaris* supplementing solid-state fermentation (CMSSF) in feed on the growth performance and immune responses of grower pigs were evaluated in the present study.

Results The *C. militaris* ethanol extract (CME) displayed potent free radical scavenging activity against 2, 2-Diphenyl-1-picrylhydrazyl (DPPH) and 2,2'-azinobis-(3-ethylbenzothiazoline-6-sulfonic acid) (ABTS) after undergoing fermentation. Additionally, the antibacterial testing revealed that CME effectively inhibits the growth of common pig pathogens such as *Glaesserella parasuis*, *Pasteurella multocida*, *Staphylococcus hyicus*, and *Streptococcus suis*. In lipopolysaccharide (LPS)-treated intestinal porcine enterocyte cell line (IPEC-J2), CME significantly suppressed the production of pro-inflammatory cytokines, including tumor necrosis factor (TNF)- α and interleukin (IL)-6. In addition, higher antioxidative activity was detected as indicated by elevated concentration of superoxide dismutase (SOD) in pig serum. The levels of immunoglobulin M (IgM), IgA, and IgG antibodies, as well as classical swine fever virus (CSFV) antibodies (S/P ratio) in serum were all increased. Growth performance of pigs fed with dietary CMSSF supplementation was improved in comparison with the control.

Conclusions Results demonstrated that CMSSF has the potential to be used as a natural growth promoter to enhance immunity, antioxidation, as well as overall health and growth performance of grower pigs.

Keywords *Cordyceps militaris* solid-state fermentation (CMSSF), Antibiotic growth promoters (AGPs), Intestinal porcine enterocyte cell line (IPEC-J2), Antibacterial, Immunity, Antioxidation

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Background

Pathogenic infections are a common, yet intricate issue encountered in livestock production. The occurrence of these infections is influenced by a variety of factors, such as seasonality, pathogen species [1], environment management [2], and biosecurity [3], which collectively make them challenging to control. The outbreak of infectious diseases attributes detrimental impacts to animal health and performance, as well as increased production costs due to medication expenses and higher mortality rates [4]. Given the increasing global concerns surrounding the use of antibiotics as growth promoters in animal feed, the exploration and implementation of safe and effective alternatives have become paramount for sustainable livestock production [5].

Traditional Chinese medicine offers a promising approach to combat antibiotic resistance. Its active components exhibit significant bacteriostatic and bactericidal effects against drug-resistant bacteria. Combined with conventional antibiotics, they show synergistic activity, potentially enhancing susceptibility, reducing dosages, and mitigating resistance development [6]. Herbs, spices, and various plant extracts are being evaluated as alternatives to antibiotics, with some showing growth-promoting effects, antimicrobial properties, and other health-related benefits [7]. Phytogetic substances, including herbs, botanicals, essential oils, and oleoresins, are a promising alternative to antibiotics in poultry feed. They improve digestion, immunity, nutrient absorption, and act as natural antimicrobials, antioxidants, and anti-inflammatory agents [8]. The study of phytochemicals has revealed potent antioxidant and anti-inflammatory activities both in vitro and in vivo [9, 10].

In general, animals naturally produce free radicals in vivo that play an important role in cell division and growth. However, excessive production of reactive oxygen species (ROS), when combined with oxo-products that do not belong to radicals, can cause damages to body tissues. Research has demonstrated that feed additives have the potential to ameliorate oxidative stress-induced injuries. For instance, feeding post-weaning piglets with dietary tea polyphenols under oxidative stress elevated the ratio of CD4⁺/CD8⁺ and helped recovery from immune damages [11]. Antioxidant nutrients, plasma proteins, and antioxidant enzymes in animals have the potential to prevent free radical-induced damages and suppress oxidative stress to maintain intestinal barrier function and free radical metabolism in post-weaning pigs [12].

Cordyceps spp. have long been considered as natural medicinal fungi in Asian medicine. Among these, *Cordyceps sinensis* is widely used in both food and medicine for its antioxidant, anti-inflammatory, antitumor [13], immune potentiation [14], and antiviral properties [15].

In addition, *C. militaris* is currently being investigated for its pharmacological functions and contains bioactive compounds such as polysaccharides, nucleosides, ergothioneine, cordycepin, and others associated with potential health benefits. These compounds demonstrate various properties, including antioxidant, antimicrobial, immunomodulatory, and anti-cancer effects [16].

Efficient harvesting of these potent bioactive molecules is achieved through water extraction, a widely recognized technique. An acidic polysaccharide was isolated from the extract of *C. militaris* grown on germinated soybeans, and when the polysaccharide was intranasally administered, it reduced virus titers in the bronchoalveolar lavage fluid and the lungs of mice infected with influenza A virus, thereby increasing the survival rate [17]. Cordycepin (3'-deoxyadenosine), the major active secondary metabolite of *C. militaris*, is an adenosine analogue that exhibits significant antimicrobial activity. Studies have shown its potent growth-inhibiting effects on *Clostridium* spp., without adverse effects on probiotics such as *Bifidobacteria* and *Lactobacilli* [18]. On the other hand, cordycepin exerts inhibitory effects on the growth of mouse melanoma and lung carcinoma cells by stimulating adenosine A3 receptors on tumor cells [19]. Cordycepin's antiproliferative effects also involve G protein activation, cAMP formation inhibition, and suppression of GSK-3 β / β -catenin signaling, resulting in decreased cyclin D1 and c-myc expression. Furthermore, it induces apoptosis by binding to DR3 receptors and activating the caspase-8/-3 pathway [20]. In vivo studies have shown that orally administered cordycepin inhibits malignant melanoma growth and exhibits antimetastatic effects in mice without adverse systemic effects [21].

Moreover, recent studies revealed that the hot water extract obtained from the fruiting bodies of *C. militaris* exhibits inhibitory effects on nitric oxide production, TNF- α , and IL-6 secretion in LPS-stimulated murine macrophage cells [22]. The inflammatory response is regulated by nitric oxide produced from macrophages through inducible nitric oxide synthase (iNOS) and cyclooxygenase-2 (COX-2) modulation. Additionally, cordycepin down-regulates iNOS and COX-2 gene expression via Akt and p38 phosphorylation suppression, leading to the inhibition of LPS-induced NO production in macrophage cells [23]. Cordycepin also enhances interleukin-10 (IL-10) expression, reduces interleukin-2 (IL-2) expression, and suppresses T lymphocyte activity [24]. Moreover, studies highlighted that feed supplemented with 1 mg/kg *C. militaris* fermentation (CMF) significantly enhances the growth performance of piglets, resulting in higher body weight gain, average daily gain, and feed intake compared to the control group [25]. The study also revealed that CMF increased the synthesis of

Th1 cytokines, such as IL-2 and IFN- γ , resulting in a substantial improvement in cell-mediated immunity.

The implementation of phytochemicals has yielded promising results, thereby prompting further exploration into their viability for integration into livestock production as a means of mitigating the proliferation of drug-resistant pathogens [8]. To this end, the present study aims to investigate the applicability of traditional herbal medicine, such as *Vigna radiata* hull, *Houttuynia cordata* stem, and wheat bran, as a fermentation substrate for solid-state fermentation using *C. militaris*. The study encompassed a range of assessments, starting from antibacterial efficacy and measuring cytokine responses in LPS-treated IPEC-J2 cells exposed to different levels of *C. militaris* extract. Additionally, the study conducts a field trial to determine the potential impact of dietary CMSSF supplementation on the growth, antioxidant, and immune responses of grower pigs. The aim is to provide evidence of the effectiveness of CMSSF as a feed additive in livestock production, contributing to the body of knowledge on the use of herbal medicine-based phytochemical treatments.

Results

Characterization of solid-state fermented *Cordyceps militaris*

The Taguchi orthogonal array design was utilized to optimize the solid-state fermentation process (Fig. 1A). The optimized traditional herbal medicine matrix consisted of 60 g wheat bran, 5 g *Houttuynia cordata* leaves, and 20 g *Vigna radiata* hulls, with 200 g moisture content in a fermentation unit. On the 14th day of fermentation, the white mycelium had fully spread on the CMSSF, covering approximately 80% of the matrix surface. A representative high-performance liquid chromatography (HPLC) chromatogram of CME showed that cordycepin and adenosine were found to be the active metabolite compounds in CME (Fig. 1B). Analysis of the cordycepin concentration from 4 to 14 days showed that group 8 had the highest cordycepin concentration on the 4th day and maintained its superiority throughout the fermentation period (Fig. 1C). The concentration of cordycepin was identified as the major compound in CME, reaching 3.8 mg/ml. Moreover, during the fermentation process, the total polyphenol content of CMSSF significantly increased from approximately 11.06 mg/g in the unfermented substrate to 15.66 mg/g on the 6th day and further to 22.22 mg/g on the 12th day (Fig. 1D). Furthermore, the bioactive components of CMSSF included not only cordycepin but also polysaccharide, which could be a potential source for prebiotics with immunomodulating properties [26]. The content of water-soluble polysaccharides significantly increased on the 12th day (Fig. 1E).

Antibacterial efficacy of CME

To assess the antibacterial efficacy of CME, bacterial growth and minimum inhibitory concentration (MIC) were determined using OD₆₀₀ measurements. MIC represents the lowest concentration of an antimicrobial agent that inhibits visible bacterial growth, used to assess bacterial susceptibility and antimicrobial efficacy [27].

In the test of growth inhibition on Gram-positive bacterial strains, *Staphylococcus hyicus* and *Streptococcus suis*, were slightly inhibited by CME at the concentration of 1000 μ g/mL, resulting in slight turbidity (Fig. 2A and B). However, at a concentration of 2000 μ g/mL, the inhibitory effect on *Staphylococcus suis* was enhanced to 100% (Fig. 2A). In contrast, CME had no inhibitory effects on none of the five Gram-negative strains (*Escherichia coli*) (Fig. 2C).

Five strains of *Pasteurella multocida* and *Glaesserella parasuis* exhibited high sensitivity to CME (Fig. 2D and E). Specifically, *Glaesserella parasuis*, when the CME was used at the concentration of 2000 μ g/mL. However, CME only had inhibitory effect on only one field strain of *Actinobacillus pleuropneumoniae* (Fig. 2F).

Antioxidant activities of CME

The scavenging ability of CME on DPPH and ABTS radicals displayed a dose-dependent pattern, as illustrated in Fig. 3. The dose-response curve obtained from these experiments enabled the determination of EC₅₀ values, measured at 23.29 mg/ml for DPPH (Fig. 3A) and 34.13 mg/ml for ABTS (Fig. 3B). CME demonstrated exceptional antioxidant activity in scavenging radicals, significantly surpassing that of the unfermented matrix extract. This enhanced activity may be attributed to the plausible augmentation of the total phenolic content by CMSSF.

Effect of CME on LPS-induced inflammatory response of porcine intestinal epithelial cells

To ensure the safety of the unfermented matrix extract and CME, we first examined the extract's effect on the viability of IPEC-J2 cells. As shown in Fig. 4A, the viability of IPEC-J2 cells was not significantly affected after incubation with 25–100 μ g/mL of CME for 24 h. Therefore, a dose of 100 μ g/mL was selected as the maximum dose to evaluate the effects of CME on LPS-induced inflammatory cytokine production in IPEC-J2 cells. The effectiveness of CMSSF in modulating the immune response and reducing excessive inflammation compared to the traditional herbal substrate was determined. In the presence of 100 μ g/mL of CME, the TNF- α concentration in the culture medium was 246.8 pg/mL, which was significantly lower than that (281.0 pg/mL, 12.17% reduction) observed with unfermented matrix extract (Fig. 4B). At 25 μ g/mL and 50 μ g/mL CME, the IL-6 concentration

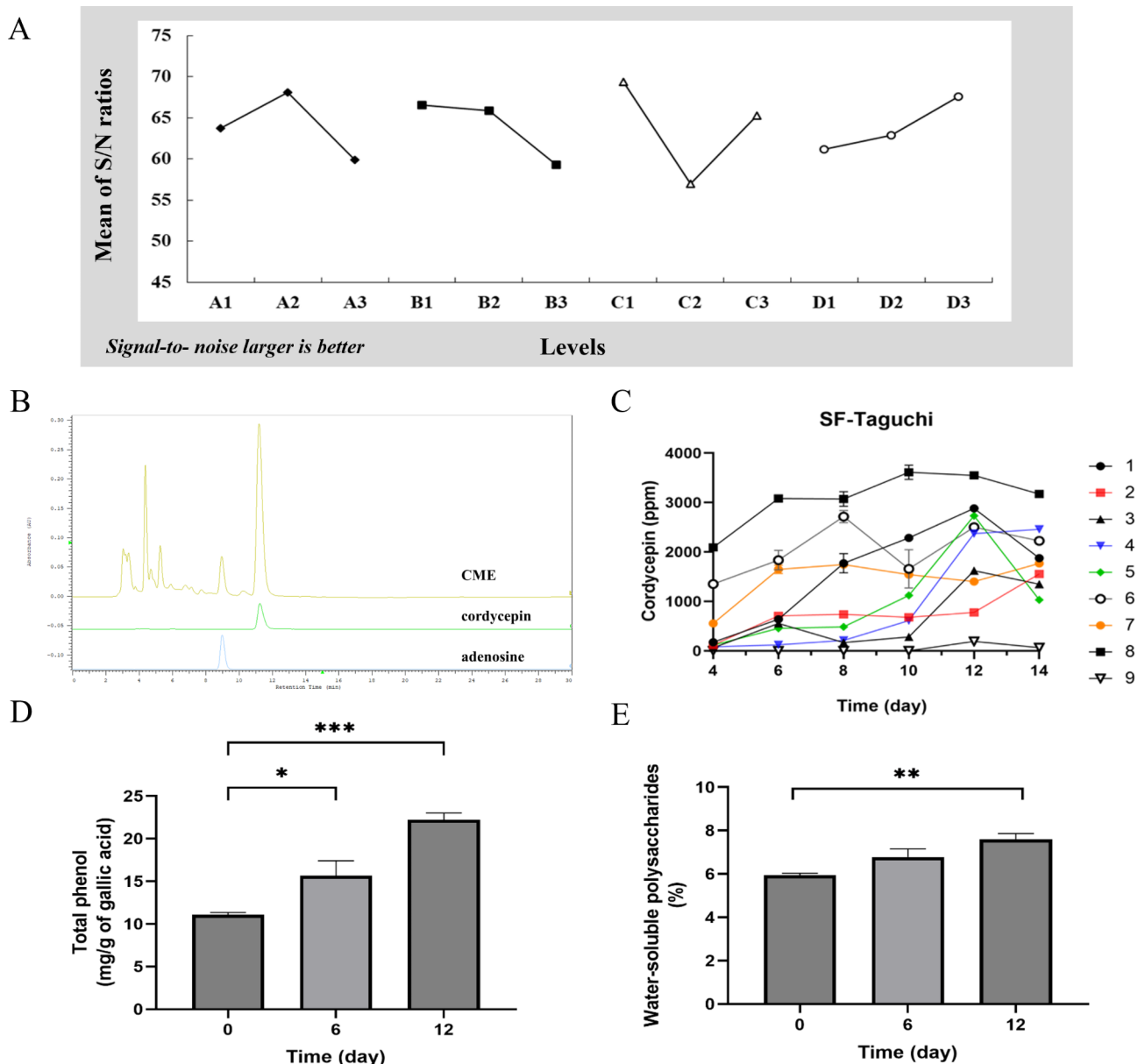


Fig. 1 The optimized cultivation factors of solid-state fermentation of *Cordyceps militaris*. **(A)** Effects of standard L9 (3^4) Taguchi orthogonal array parameters S/N ratios. **(B)** Measurement of cordycepin by HPLC method. **(C)** The changes of cordycepin content during the growth of *Cordyceps militaris* in solid-state fermentation in the Taguchi experiment. **(D)** The total polyphenol content. **(E)** The water-soluble polysaccharides content. Each value is expressed as mean $\pm$ SE ($n=3$). The p-values of < 0.05 were regarded as statistically significant, expressed as * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

was 21.8 $\mu\text{g/mL}$ and 4.6 $\mu\text{g/mL}$, respectively, which was significantly ($p < 0.05$) decreased in comparison with the unfermented matrix extract (Fig. 4C). However, at 100 $\mu\text{g/mL}$ CME, LPS-induced IL-6 production was completely inhibited. The results demonstrated that CME reduced TNF- α and IL-6 expression in a dose-dependent manner, suggesting its potential to suppress LPS-stimulated cytokine production in IPEC-J2 cells.

Growth performance of pigs fed diets supplemented with CMSSF

The effects of dietary supplemented CMSSF on the growth performance of grower pigs are presented in Table 1. The inclusion of CMSSF had no significant influence on average daily gain (ADG), average daily feed intake (ADFI), or feed efficiency over the 8-week feeding period. However, supplementation with 5 g/kg CMSSF in the early feeding period 0–4 weeks and 1 g/kg CMSSF in the late feeding period (5–8 weeks) improved the feed conversion rate compared to the control group and

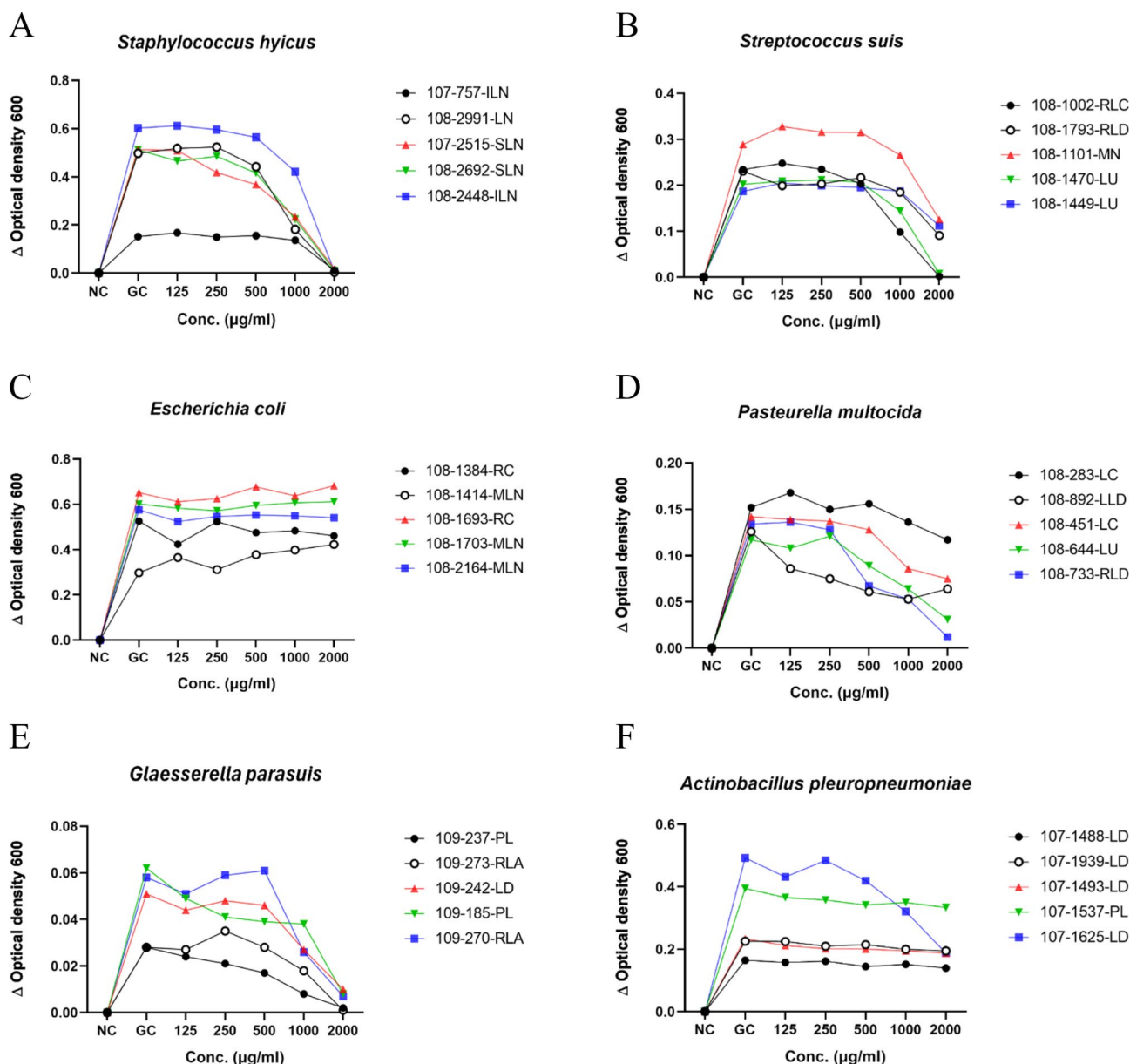


Fig. 2 Antimicrobial susceptibility testing of ethanol extract of *Cordyceps militaris* in solid-state fermentation. NC is negative control without bacteria in the medium and GC is positive control with bacteria growth. Gram (+) bacterial strains include (A) *Staphylococcus hyicus*, (B) *Streptococcus suis*. Gram (-) bacterial strains include (C) *Escherichia coli*, (D) *Pasteurella multocida*, (E) *Glaesserella parasuis*, and (F) *Actinobacillus pleuropneumoniae*. Bacterial growth and MIC were observed by OD₆₀₀ measurement

antibiotic group. The dietary addition of a higher dose of 20 g/kg CMSSF did not affect the survival of growing pigs, which remained at 100% (data not shown). These results indicate that CMSSF exhibits no adverse reactions or toxicity in grower pigs.

Antibody levels of pigs fed diets supplemented with CMSSF

Serum immunoglobulin levels (IgA, IgG, IgM) were assessed to determine the influence of CMSSF on pig's immunity. At the onset of the experiment, the initial

concentrations of immunoglobulins indicated no significant differences among the groups (Fig. 5). However, by the 4th week, a noticeable rise in IgA was observed with the administration of 5 g/kg CMSSE, but no statistically significant differences were observed among the levels of IgA, IgM, and IgG levels overall (Fig. 5A). Significant differences were observed at the 8th week after administering 1 g/kg of CMSSE. The levels of IgA, IgM, and IgG significantly ($p < 0.05$) increased (by 20.93%, 46.64%, and 60.97%, respectively) when compared to the control group (Fig. 5A-C). Interestingly, higher doses of CMSSF

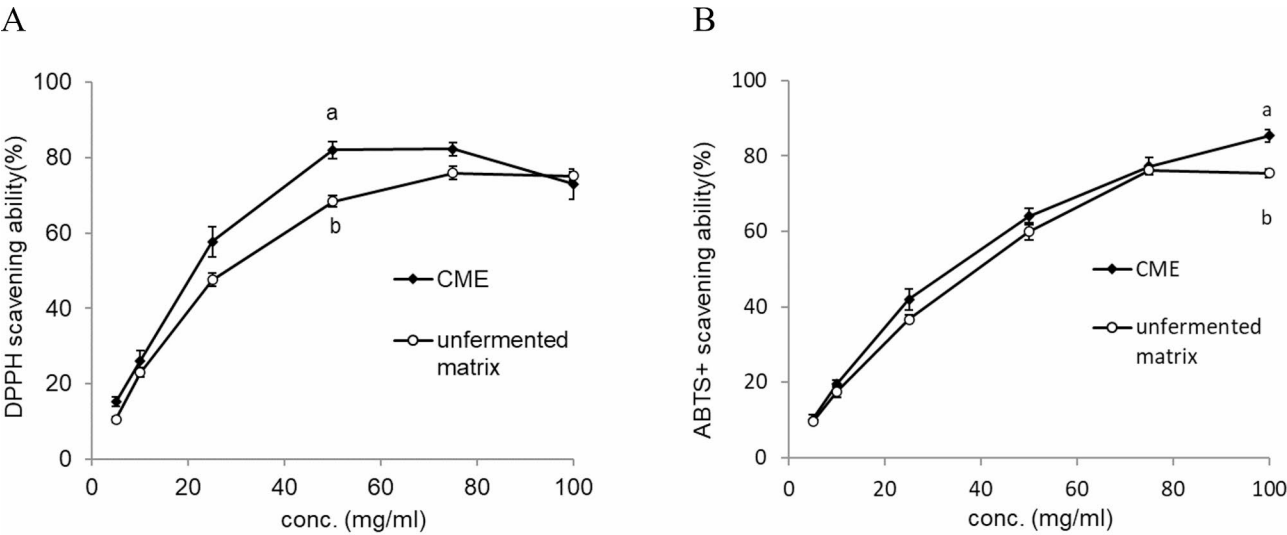


Fig. 3 Antioxidant activities of ethanol extract from solid-state fermentation of *Cordyceps militaris*. **(A)** DPPH scavenging activity **(B)** ABTS scavenging activity is compared with the activity of 10mM Trolox values attained. EC50 is the efficient concentration of the test samples, which decreases the initial radical by 50%. It was obtained by interpolation from linear regression analysis. Each value is expressed as mean ± SE (n=3). * p < 0.05

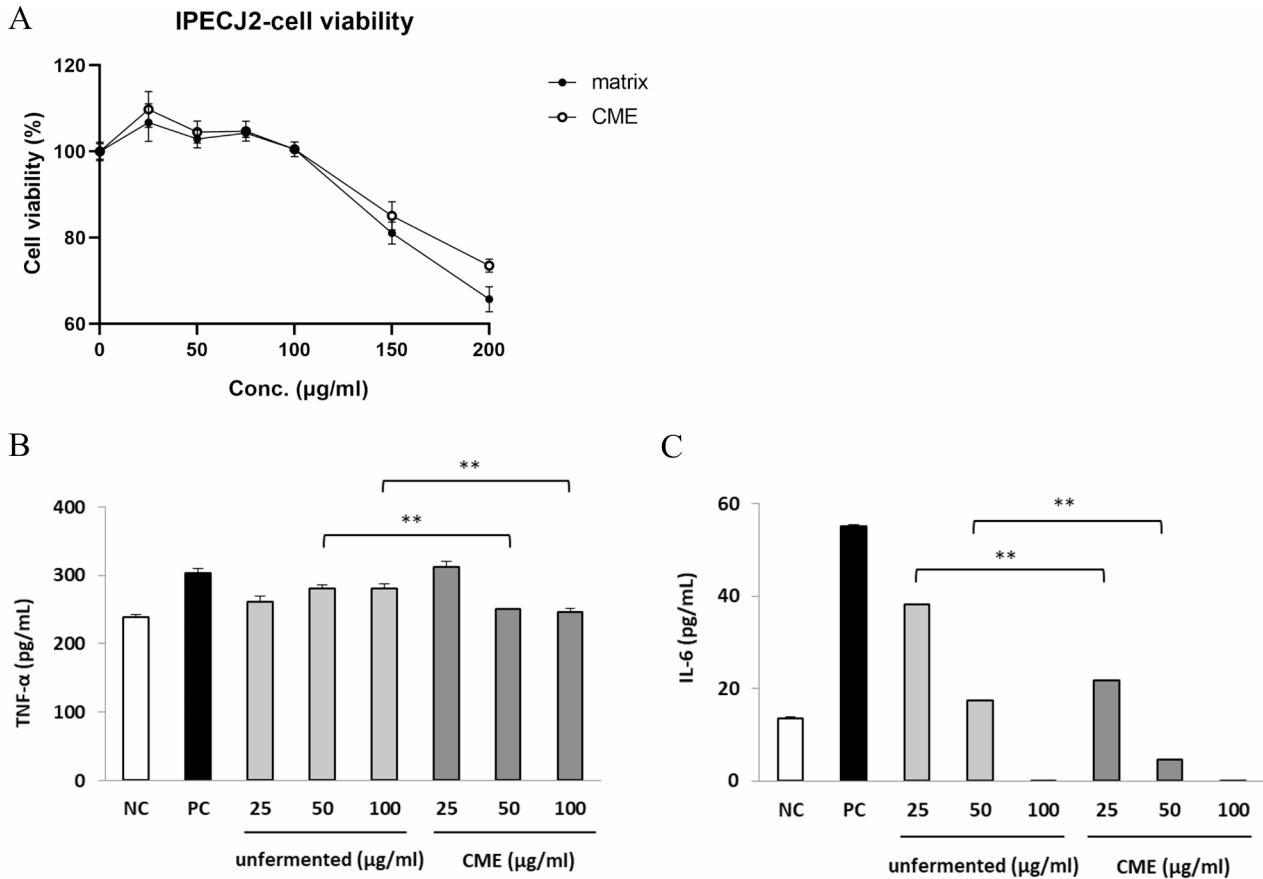


Fig. 4 The inflammation cytokine production of unfermented matrix extract and ethanol extract of *Cordyceps militaris* solid-state fermentation in the IPEC-J2. **(A)** Cell viability was estimated with the indicated concentrations of extracts by MTT assay for 24 h. **(B)** Analysis of LPS induced TNF-α production by ELISA. Cells were treated with the indicated concentrations of extract in the presence of LPS or LPS alone for 24 h. **(C)** The IL-6 production was determined by ELISA. Cells were treated with the indicated concentrations of extract in the presence of LPS or LPS alone for 24 h. Each value is expressed as mean ± SE (n=3). ** p < 0.01

Table 1 Effect of CMSSF on body weight, food intake and feed conversion rate of grower pigs

Items	Control	PC	L	M	H	SEM	P-value
Body Weight (kg)							
0 week	26.7	26.71	26.73	26.73	26.73	0.03	1
4 week	47.94	48.71	47.54	48.77	47.81	0.22	0.28
8 week	73.92	74.52	72.69	74.31	72.56	0.35	0.25
Average Daily Gain (kg)							
0–4 week	0.759	0.786	0.743	0.787	0.753	0.01	0.32
5–8 week	0.928	0.922	0.898	0.912	0.884	0.01	0.8
0–8 week	0.843	0.854	0.821	0.85	0.818	0.01	0.29
Feed Intake (kg/day)							
0–4 week	1.469	1.51	1.471	1.487	1.485	0.02	0.98
5–8 week	2.076	2.053	1.942	2.03	1.979	0.03	0.51
0–8 week	1.773	1.781	1.707	1.758	1.732	0.02	0.64
Feed Conversion Rate							
0–4 week	1.94	1.92	1.98	1.89	1.97	0.02	0.72
5–8 week	2.24	2.23	2.16	2.23	2.24	0.02	0.79
0–8 week	2.1	2.09	2.08	2.07	2.12	0.02	0.92

Control: general feed, L: 1 g CMSSF/kg feed, M: 5 g CMSSF/kg feed, H: 20 g CMSSF/kg feed, PC: Tiamulin 100 ppm + Oxytetracycline 220 ppm

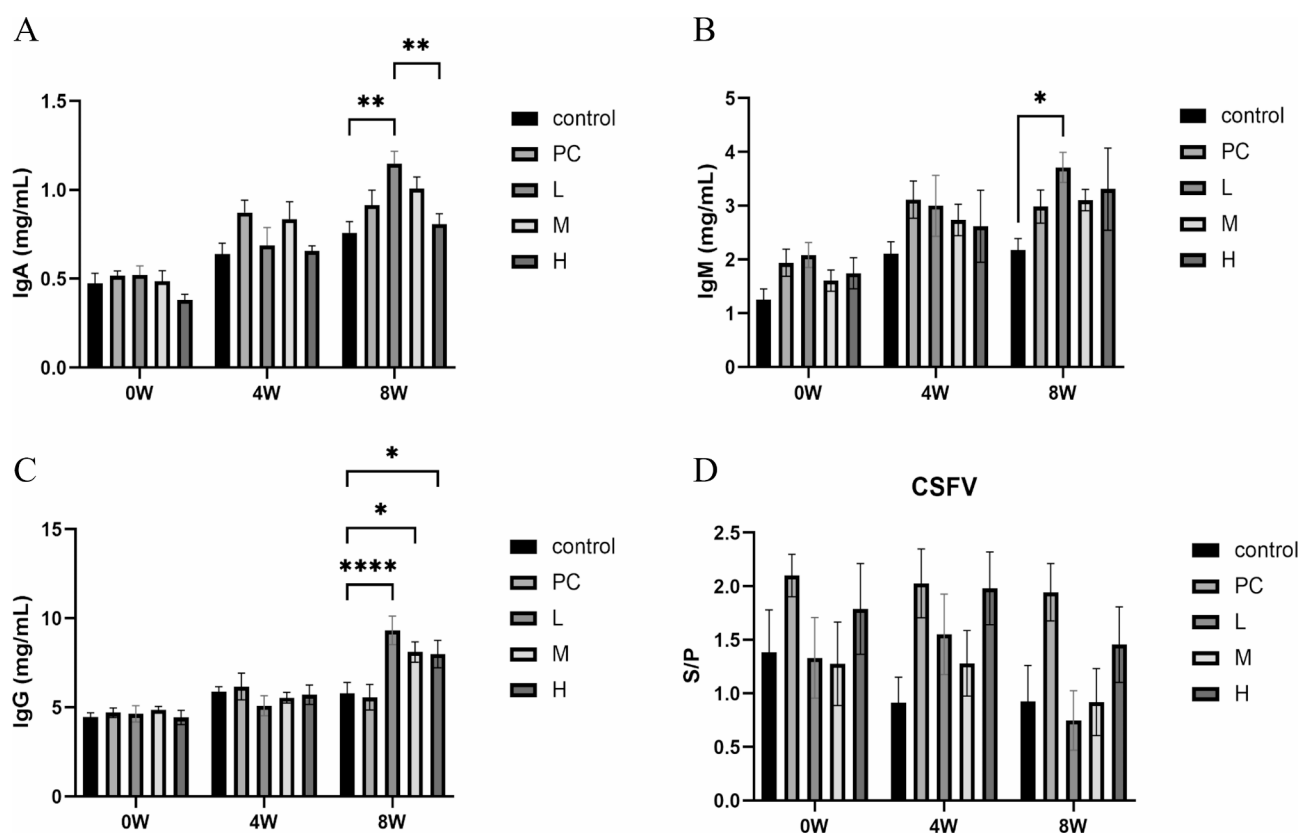


Fig. 5 The effect of CMSSF supplementation levels on swine serum Immunoglobulins (A) IgA, (B) IgM, (C) IgG concentrations via ELISA and (D) CSFV antibody level (S/P Ratio). Control general feed, L: 1 g CMSSF/kg feed, M: 5 g CMSSF/kg feed, H: 20 g CMSSF/kg feed, PC: Tiamulin 100 ppm + Oxytetracycline 220 ppm. The p-values of < 0.05 were regarded as statistically significant, expressed as * $p < 0.05$; ** $p < 0.01$; **** $p < 0.0001$

had the opposite effect, leading to reduced concentrations of IgA, IgG, and IgM.

Following the analysis of CSFV antibody levels utilizing the S/P ratio, it was determined that all doses of CMSSF were effective in increasing CSFV-specific antibody

levels by the 4th week. Among all the groups, the 20 g/kg CMSSF group had the highest level of antibody against CSFV. By the 8th week, dietary supplementation of CMSSF increased CSFV-specific antibody in a dose-dependent manner. The antibiotic group consistently

demonstrated increased antibody levels at both time points (Fig. 5D).

Plasma and erythrocyte antioxidant capacity of grower pigs fed diets supplemented with CMSSF

The impacts of an 8-week dietary supplementation of CMSSF on the plasma and erythrocyte antioxidant capacities of growing pigs are presented in Table 2. The findings revealed that diets supplemented with CMSSF had no significant impact on plasma trolox equivalent antioxidant capacity (TEAC) and whole blood glutathione peroxidase (GPx) activity compared to the control. However, grower pigs fed diets containing 1 g/kg CMSSF had increased erythrocyte SOD, from 2840 ± 567.9 (U/g, Hb) to 4035 ± 815.0 (U/g, Hb), after 8 weeks. Additionally, the highest increase was observed when the CMSSF inclusion rate was 20 g/kg, reaching 5095 ± 1317.7 (U/g, Hb). Furthermore, the inclusion of CMSSF in the feed significantly ($p < 0.05$) enhances erythrocyte SOD activity in a dose-dependent manner when compared to the control group.

Discussion

The development and application of potential antibiotic-alternative phytochemicals in commercial settings are crucial for reducing reliance on AGPs and establishing a sustainable animal production system in the absence of antibiotics [28]. In this study, we investigated the utilization of *Vigna radiata* hull, *Houttuynia cordata* stem, and wheat bran as substrates for *Cordyceps* fermentation for in vivo application in growing pigs. Our research focused on evaluating the antibacterial, antioxidant, and anti-inflammatory properties of *C. militaris* solid-state fermented phytogenic feed additive. In previous studies, fermented *Vigna radiata* hull was found to enhance the antioxidant and immune responses of weaned pigs [29], whereas the *Vigna radiata* ethanol extract exhibited antiviral activity by targeting viral attachment, penetration, assembly, and release to inhibit Influenza A virus [30]. The dietary supplementation of *H. cordata* was observed

to reduce the fecal *E. coli* concentration in weaning pigs [31], while it also contributed to improved growth performance and a reduced incidence of diarrhea in pigs, along with an increase in the longissimus muscle area in finishing pigs [32].

After weaning, pigs often experience intestinal dysfunction that may lead to an imbalanced status in the antioxidant system, which affects gut health. Antioxidants play a crucial role in restoring gut health and reducing the stress associated with weaning in post-weaning pigs [12]. Antioxidant supplementation provides direct benefits by eliminating excess free radicals and promoting antioxidative effects [33]. The design of the grower pig experiment is based on addressing intestinal dysfunction and an imbalance in the antioxidant system following weaning. Weaning-associated stress also makes piglets highly susceptible to bacterial infections in the gastrointestinal tract, which is why antibiotics are widely used during this phase to mitigate these effects. In contrast, grower pigs (20–60 kg) are more prone to viral respiratory infections and relied on antibiotic administration less. Thus, we thought using the grower pigs (20–60 kg) would be a more suitable phase for investigating antibiotic alternatives, such as *Cordyceps* fermentation products. The inclusion of a positive control group receiving 100 ppm of Tiamulin and 220 ppm of Oxytetracycline allowed for a direct comparison between standard antibiotic administration and the alternative CMSSF.

We assessed overall blood antioxidant levels through plasma TEAC measurements. Additionally, we measured the activity of enzymes such as SOD and GPx, which serve as indicators of cellular metabolic responses to free radicals [34]. The methanolic extract of *C. militaris* demonstrated the ability to inhibit lipid peroxidation, scavenge free radicals, and exhibit strong antibacterial and antifungal properties [35]. In our in vitro experiments, using the ethanol extract of CMSSF showed a significant increase in DPPH and ABTS+scavenging activity compared to unfermented substrate. This finding is in line with previous research suggesting that phenolic

Table 2 Effect of CMSSF on TEAC, GPx and SOD of anti-oxidation function in grower pigs

Items	Control	PC	L	M	H	SEM	P-value
TEAC (μM)							
0 Week	0.46 ^{ab} ± 0.01	0.45 ^{abc} ± 0.04	0.46 ^a ± 0.03	0.44 ^{bc} ± 0.03	0.43 ^c ± 0.05	0.01	0.05
8 Week	0.39 ± 0.05	0.39 ± 0.05	0.37 ± 0.09	0.39 ± 0.04	0.35 ± 0.05	0.02	0.23
GPx (U/mL)							
0 Week	1.65 ± 0.12	1.73 ± 0.35	1.65 ± 0.21	1.62 ± 0.20	1.63 ± 0.27	0.08	0.88
8 Week	1.78 ± 0.25	1.92 ± 0.20	1.88 ± 0.32	1.77 ± 0.22	1.69 ± 0.29	0.09	0.10
SOD (U/g, Hb)							
0 Week	3770 ± 467.2	2857 ± 705.3	2840 ± 567.9	2537 ± 383.9	3266 ± 1562.2	212.3	0.21
8 Week	3770 ^c ± 738.0	3275 ^c ± 679.9	4035 ^b ± 815.0	4426 ^{ab} ± 1023	5095 ^a ± 1317.7	237.5	0.01

a, b, c Means with different superscripts indicate significantly different among treatments, respectively ($p < 0.05$). Control: general feed, L: 1 g CMSSF/kg feed, M: 5 g CMSSF/kg feed, H: 20 g CMSSF/kg feed, PC: Tiamulin 100 ppm + Oxytetracycline 220 ppm

compounds, containing hydroxyl groups, facilitate free radical scavenging [36]. Previous studies have suggested that the potency of radical scavenging activities in extracts is associated with increased phenolic compound levels, highlighting the significance of total phenolic content in influencing the inhibition of ABTS^{•+} and DPPH[•] radicals [37]. These findings support the observed differences in antioxidant capabilities between CME and the matrix extract and emphasize the importance of total phenolic content in enhancing the antioxidant activity of extracts. In addition, it has been demonstrated that the fruiting bodies have a stronger DPPH[•] radical scavenging activity, while fermented mycelia have a stronger total antioxidant capacity, chelating ability, and reducing power, indicating that fruiting bodies and mycelia have their unique roles and work differently [38]. For this reason, we used the fermented mycelium of *C. militaris*, which is not only cost-effective but also has superior antioxidant capabilities as a functional feed additive. Additionally, diets that included CMSSF tended to improve SOD activities in erythrocytes, aiding in the clearance of free radicals. The antioxidant activity of CMSSF was evident not only in in vitro experiments but also in the growing pig model. Thus, CMSSF is a potent natural fermentation with significant antioxidant properties. Serum immunoglobulins are crucial indicators of humoral immunity in pigs. IgA and IgG play key roles in defense against pathogenic invasion, and changes in these protein levels can influence pig growth performance and immunity [39]. *Cordyceps militaris* spent mushroom substrate (CMS) supplementation in growing pigs significantly increased IgA and IgG levels, and enhanced total antioxidant capacity and GPx activity. However, feed efficiency and certain metabolic parameters remained unaffected [40]. These immunomodulatory and antioxidative effects are comparable to those observed with CMSSE, suggesting similar bioactive properties. Interestingly, while CMS enhanced total antioxidant capacity and GPx activity, CMSSF specifically increased SOD concentration, indicating potential differences in their antioxidant mechanisms.

Previous studies have illustrated the dual-action mechanism of cordycepin, which effectively restrains the growth of *E. coli* and *B. subtilis* by compromising their cytoplasmic membranes. Further evaluation revealed compromised outer membrane permeability, leading to the leakage of cytoplasmic content, along with structural disruption caused by cordycepin's insertion into genomic DNA [41]. This study used that the ethanol extract derived from *C. militaris* fermentation demonstrates significant antibacterial properties against a spectrum of pathogenic bacteria, including Gram-positive (*Staphylococcus hyicus*, *Streptococcus suis*) and Gram-negative strains (*Escherichia coli*, *Pasteurella multocida*,

Glaesserella parasuis, and *Actinobacillus pleuropneumoniae*) isolated from pigs in livestock farms. Although CME contains 3.8 mg/ml of cordycepin, it did not inhibit the growth of *E. coli* field isolates in the antibacterial efficacy test. This lack of inhibition may be attributed to the lower concentration of CME used (2000 µg/ml) compared to the effective cordycepin concentration (2500 µg/ml) reported in the literature to inhibit *E. coli* growth [41]. To further investigate this, supplementary tests were conducted using a standard *E. coli* strain (BCRC11509). The results indicated that CME inhibits *E. coli* growth only at concentrations exceeding 5000 µg/ml (data not shown). This finding suggests that higher concentrations of CME are necessary to achieve an antibacterial effect against *E. coli* compared to purified cordycepin. The differential susceptibility of bacteria to CME may be attributed to variations in cell wall architecture, resistance mechanisms, or distinct metabolic pathways across species, which affect its antibacterial efficacy. In particular, cell wall composition plays a critical role, with factors such as cell wall thickness, the presence or absence of outer membranes, and differences in wall polarity contributing to the observed variability in bacterial responses [42]. Structural differences between Gram-positive and Gram-negative bacteria significantly affect their susceptibility to CME. Gram-positive bacteria, characterized by their thicker peptidoglycan layers and lipoteichoic acid (LTA) complexes, which are composed of glycerophosphoric acid chains covalently bonded to phospholipids in the cell membrane [43], are more susceptible to hydrophobic compounds present in herbal substrates and *Cordyceps militaris* fermented extract. In contrast, Gram-negative bacteria possess an outer membrane that provides an additional barrier to external agents [42]. Nonetheless, CME can inhibit the growth of Gram-negative bacteria at higher concentrations by altering membrane permeability and inducing nucleic acid damage. The interaction between these structural and functional features largely dictates the susceptibility of various bacterial species to CME, accounting for the observed differences in antibacterial efficacy across strains, species, and underscoring its potential as a broad-spectrum antibacterial agent.

C. militaris water extract induces the phenotypic and functional maturation of dendritic cells. The water extract promotes the cytotoxicity of specific cytotoxic T lymphocytes induced by dendritic cells and enhances the immunoregulatory function in patients with impaired host defense [44]. Recent studies emphasize the crucial role of cytokines in modulating the physiological and pathological aspects of the intestinal tight junction barrier. Their pro-inflammatory activities disrupt this barrier, leading to increased penetration of luminal antigens [45]. In vivo, *Cordyceps militaris* supplementation in piglets significantly increased the villus height and villus

height to crypt depth ratio in both the duodenum and jejunum. Furthermore, *C. militaris* supplementation upregulated the expression of tight-junction proteins in these intestinal segments [46]. In the context of solid-state fermentation utilizing wheat-based substrates, the CMHW containing cordycepin demonstrates its ability to counteract the pro-inflammatory effects induced by the wheat matrix [47]. Treatment with CMHW inhibited LPS-induced phosphorylation of p38 in porcine alveolar macrophages, leading to reduced secretion of pro-inflammatory cytokines such as TNF- α , IL-1 β , and IL-6. Additionally, CMHW dose-dependently reduced LPS-stimulated nitric oxide production and expression levels of cyclooxygenase-2 protein.

While various studies have investigated the immunomodulatory effects of *C. militaris* on various cells, research specifically focusing on porcine intestinal epithelial cells remains limited. Intestinal epithelial barrier dysfunction and inflammatory responses are common pathological processes in intestinal diseases. The IPEC-J2 cells represent porcine intestinal columnar epithelial cells isolated from neonatal piglets' mid-jejunum, which hold significance due to their non-transformed nature [48]. The IPEC-J2 cell line is an excellent tool for studying the interactions between epithelial cells and enteric bacteria, evaluating the impact of probiotic microorganisms, and assessing the effects of nutrients on various parameters that reflect epithelial functionality [49]. This study employed wheat as the primary substrate and traditional herbal medicine for CMSSE. The results demonstrate that in the IPEC-J2 cell model, *C. militaris* extract significantly inhibits the LPS-induced production of the inflammatory cytokines TNF- α and IL-6 compared to the unfermented matrix. This outcome highlights the potential of *C. militaris* in reducing inflammatory responses within the specific context of porcine intestinal epithelial cells, as CMSSE increases the content of total polyphenols, water-soluble crude polysaccharides, adenosine, cordycepin, and other metabolites.

Previous research on *C. sinensis* polysaccharides has demonstrated its significant promotion of microbial-derived butyrate, which helps balance Treg/Th17 and curb TLR-mediated inflammation, along with boosting sIgA secretion to suppress colonic inflammation [50]. Furthermore, *C. sinensis* polysaccharides restored short-chain fatty acids (SCFA) levels that were reduced due to cyclophosphamide-induced mucosal immunosuppression and microbial dysbiosis in mice, thereby enhancing microbial diversity and restructuring gut microbiota. Taxonomic analysis revealed an increase in the abundance of beneficial bacteria such as *Lactobacillus*, *Bifidobacterium*, and *Bacteroides*, with a reduction in pathogenic bacteria like *Clostridium* and *Flexispira* [51]. *Cordyceps militaris* supplementation in piglets

enhanced antioxidant activity, elevated SCFA concentrations (specifically acetic and butyric acids), and increased the abundance of *Bacillus* and *Lactobacillus* species in the gut microbiota [46]. Similarly, adding 300 mg/kg *C. militaris* to the finishing pig diet notably improves gut structure, enhancing goblet cells and lymphocytes, upregulating barrier-related protein expression, reducing proinflammatory cytokines (IFN- γ , TNF- α , and IL-12), and increasing anti-inflammatory cytokines (IL-10 and IL-4) [52]. Mechanistically, *C. militaris* was found to suppress TLR4/MyD88/NF- κ B pathway proteins, impacting gut microbes and elevating acetate and butyrate, similar to the effects of *C. sinensis* polysaccharides [52]. These effects were consistent with our findings in in vitro and growing pigs, suggesting a common mechanism of action across porcine developmental stages, likely mediated through improved antioxidant capacity and intestinal epithelial function.

Conclusion

This study investigated the synergistic effects of *Cordyceps militaris* fermentation with herbal substrates on grower pig health. The fermentation process augmented the bioactive properties of both *C. militaris* and the herbal matrix, yielding CMSSE with enhanced antioxidant and immunomodulatory efficacy. Dietary CMSSE significantly improved antioxidative status, elevated immunoglobulin levels, and enhanced erythrocyte antioxidant capacity in grower pigs, notably through increased superoxide dismutase activity. These effects exhibited a positive dose-response relationship, highlighting the potency of this synergistic approach. Moreover, the CME, abundant in polyphenols and polysaccharides, demonstrated broad-spectrum antimicrobial activity against both Gram-positive and Gram-negative bacteria, albeit at varying minimum inhibitory concentrations. In vitro analyses further revealed CME's capacity to attenuate gut inflammatory responses. These findings suggest that CMSSE, harnessing the synergistic fermentation of *C. militaris* and herbal substrates, may serve as a natural, broad-spectrum alternative to antibiotic growth promoters, potentially enhancing both the health status and performance parameters of grower pigs.

Methods

Cordyceps militaris solid-state fermentation and CME preparation

The *C. militaris* strain (BCRC33740) was initially subcultured onto potato dextrose agar (BD Difco™) and incubated at 25 °C for 14 days. Subsequently, *C. militaris* was inoculated into potato dextrose broth (BD Difco™) at 25 °C with agitation at 130 rpm for 5 days.

A Taguchi orthogonal array design was utilized for the solid-state fermentation process. Four key factors,

namely wheat bran, *Houttuynia cordata* leaves, *Vigna radiata* hulls, and moisture content, were selected for optimization due to their established roles in promoting microbial growth and metabolite production in solid-state fermentation systems. Each factor was varied at three levels to thoroughly investigate the fermentation process. The experimental matrix was generated using the Taguchi L9 (3⁴) orthogonal array (Table 3) to minimize the number of required experimental runs while ensuring robustness and efficiency. The substrates were autoclaved, inoculated with 10% (v/w) of the inoculum under sterile conditions, and then incubated in a dark incubator at 25 °C for 14 days. The crude fermentation product was dried at 50 °C to reduce the moisture content (<8%), followed by grinding into powders to obtain CMSSF. The CMSSF was subsequently extracted using 50% ethanol (1:5 w/v) and sonicated for 1 h. After centrifugation at 8,000×g for 10 min, the supernatant was concentrated using a rotary evaporator (Eyela, Japan) and freeze-dried to obtain CME. The recovery of dried CME was approximately 18–26% (w/w), and the dried CME was stored at −20 °C until use.

The data obtained from the experiments were analyzed using Taguchi’s signal-to-noise (S/N) ratio analysis and analysis of variance (ANOVA).

Table 3 Taguchi-based design L9 (3⁴): 4 factors, 3 levels.

Levels	Factors			
	A	B	C	D
	wheat bran (g)	<i>Houttuynia cordata</i> leaves (g)	<i>Vigna radiata</i> hulls (g)	Moisture (g)
L1	40	5	20	100
L2	60	10	50	150
L3	80	20	70	200

Extraction and analysis of water-soluble crude polysaccharides

The CMSSF was extracted with ddH₂O (1:10 w/v) and autoclaved for 20 min, followed by filtration through a Whatman No. 1 filter. The amount of water-soluble crude polysaccharides of *C. militaris* was analyzed by treating the filtrate of the hot water extract with α-amylase to digest matrix polysaccharides at 50 °C for 1 h. After the enzyme reaction, the *C. militaris* hot water (CMHW) extract was dialyzed using a Spectra/Por1 dialysis membrane (MWCO 6–8 K, Repligen) and incubated at 4 °C overnight to ensure full retention of water-soluble crude polysaccharides with a molecular weight>6 kD. The concentration of polysaccharides was determined by OD490 phenol-sulfuric acid quantification method [53].

Analysis of the total polyphenolic content

The amount of total phenolics in CME was estimated using the Folin-Ciocalteu method [54]. An appropriate volume of the CME or FeSO₄ solution (30 μL) was added into a 1.5mL microcentrifuge tube, mixed with ddH₂O (700 μL) and 1 N Follin-Ciocalteu reagent (150 μL, previously diluted 1: 2 (v/v) with ddH₂O). After 5 min at 25 °C, a solution of Na₂CO₃ (120 μL, 15%, w/w) was added, and the sample was mixed again and incubated for an additional 2 h. The absorbance of supernatants at 765 nm was measured with UV/VIS spectrometer (BioTek, U.S.). All tests were performed in triplicate and results are expressed as milligrams of gallic acid equivalents per gram of dry weight (mg GAE/g) of extract.

HPLC analysis of cordycepin

The analysis of CMSSF was performed using an HPLC system (Hitachi D2000, Tokyo, Japan), which consisted of an L-2130 HTA pump, an L-2200 autosampler, and an L-2455 photodiode array detector. All solvents used were HPLC grade, and all reagents were analytical grade. The extracted compounds from CME, the adenosine and cordycepin standard (Sigma-Aldrich, St. Louis, MO, USA) were separated using an Inertsil ODS-2 C18 column (250 mm × 4.6 mm, 5 μm) with a mobile phase of methanol (solvent A) and ddH₂O (solvent B). The mobile phase consisted of methanol: ddH₂O (17:83, v/v). The flow rate was maintained at 0.8 mL/min with an injection volume of 10 μL. The column temperature was maintained at 30 °C, and the absorbance of the eluate was continuously monitored at 260 nm. Chromatographic peaks were identified by spiking the samples with standard compounds, and their UV spectra and retention times were matched with standard compounds.

In vitro antioxidant assays

DPPH and ABTS radical scavenging assay were conducted according to a modified method [54, 55]. Briefly, a 0.4 mM DPPH solution in methanol was vigorously mixed with various concentrations (5–100 mg/ml in 60% ethanol) of CME. After a 90-min incubation in darkness at room temperature, the supernatant was measured for absorbance at 517 nm using a UV/VIS spectrometer (BioTek, U.S.). To prepare the ABTS+solution, 7 mM ABTS was mixed with 140 mM potassium peroxydisulfate and allowed to react in the dark 16 h to form blue-green ABTS+. The ABTS+solution was diluted with phosphate-buffered saline (pH 7.4) to a final absorbance of 0.70±0.02 at 734 nm. ABTS+solution was mixed with CME, and the absorbance at 734 nm was measured in time.

The percentage of free radical scavenging ability (mg powder/mL) was calculated using the following equation:

$$\text{Scavenging ability (\%)} = \left[1 - \left(\frac{A_{\text{sample}}}{A_{\text{cont}}} \right) \right] \times 100\% \quad (1)$$

Where A_{cont} and A_{sample} represent the absorbance values for the control and sample, respectively. Trolox was observed to have a small impact on the number of exchanged electrons, confirming its suitability as a positive control standard compound in comparison to other antioxidants [56].

Antimicrobial efficacy test

The broth dilution method was used to determine the MIC value, which assesses bacterial susceptibility to antimicrobial agents and evaluates the activity of new compounds. Bacteria were inoculated into a liquid growth medium containing varying concentrations of the antimicrobial agent. After a specified incubation period, bacterial growth was assessed, and the MIC was determined as the lowest concentration of the agent that resulted in a clear bacterial suspension, indicating inhibition of growth [27]. This experiment focused on testing the antibacterial efficacy of five strains belonging to different bacterial species, including *Staphylococcus hyicus* (107-757-ILN, 108-2991-LN, 107-2515-SLN, 108-2692-SLN, 108-2448-ILN), *Streptococcus suis* (108-1002-RLC, 108-1793-RLD, 108-1101-MN, 108-1470-LU, 108-1449-LU), *Escherichia coli* (108-1384-RC, 108-1414-MLN, 108-1693-RC, 108-1703-MLN, 108-2164-MLN), *Pasteurella multocida* (108-283-LC, 108-892-LLD, 108-451-LC, 108-644-LU, 108-733-RLD), *Glaesserella parasuis* (109-237-PL, 109-273-RLA, 109-242-LD, 109-185-PL, 109-270-RLA), and *Actinobacillus pleuropneumoniae* (107-1488-LD, 107-1939-LD, 107-1493-LD, 107-1537-PL, 107-1625-LD). All strains were isolated from the livestock farm and donated by Dr. Ming-Tang Chiou and Dr. Chao-Nan Lin (Animal Disease Diagnostic Center, College of Veterinary Medicine, National Pingtung University of Science and Technology, Taiwan), and exhibited higher susceptibility to pig disease compared to standard strains. The culture conditions for each strain were as follows: *Staphylococcus hyicus*, *Streptococcus suis*, *Escherichia coli*, and *Pasteurella multocida* were cultured on blood agar containing 5% sheep blood at 37 °C for 16–18 h. *Glaesserella parasuis* was cultured on self-prepared chocolate agar at 37 °C in a 5% CO₂ atmosphere for 18–24 h. *Actinobacillus pleuropneumoniae* was cultured on chocolate agar at 37 °C in a 5% CO₂ atmosphere for 16–18 h.

For *Staphylococcus hyicus*, *Escherichia coli*, and *Pasteurella multocida*, Mueller-Hinton broth (MH) was used. *Streptococcus suis* and *Glaesserella parasuis* were tested using a modified formulation of MH liquid culture media. *Actinobacillus pleuropneumoniae* was tested using a modified Brain Heart Infusion broth (BHI) formulation. The bacterial concentrations were adjusted to

10⁶ colony forming units/mL (CFU/mL) based on linear equations correlating optical density at 600 nm (OD₆₀₀) with CFU/mL. The final bacterial concentration was 5×10⁵ CFU/mL, and the indicated CME concentration was mixed and incubated under appropriate conditions for 24 h. Respectively, liquid culture media without CME served as the growth control (GC) and negative control (NC) groups. OD₆₀₀ measurements were recorded for subsequent analysis.

Cell culture and inflammatory cytokine detection

IPEC-J2 cells were kindly provided by Dr. Je-Ruei Liu (Institute of Biotechnology, National Taiwan University, Taiwan). Cells were cultured in complete growth medium, consisting of Dulbecco's Modified Eagle's Medium/high glucose (DMEM) medium supplemented with 10% (v/v) fetal bovine serum and 1% PSA (Penicillin / Streptomycin / Amphotericin B) at 37°C in humidified air with 5% (v/v) CO₂. The IPEC-J2 cells were seeded into 96-well plates at a density of 3×10⁴ cells/well and treated with the indicated concentrations of CME for 24 h. Cell viability was assessed using the 3-(4,5-dimethylthiazol-2-yl)-2,5 diphenyl tetrazolium bromide (MTT) assay as previously described [57], the inflammatory cytokines in IPEC-J2 were determined using the porcine IL-6, TNF-α ELISA Kit (MyBioSource, San Diego, USA) [58]. Following induction of inflammation with 40 µg/mL LPS for 15 h, cell culture supernatants were collected and analyzed using a spectrometer (BioTek, U.S.).

Growth performance of grower pigs

A total of 80 healthy cross-bred (Landrace × Yorkshire × Duroc) grower pigs (weighing 25±2 kg) were used in our study. They were sourced from the commercial farm supervised by the Agricultural Technology Research Institute, Taiwan and randomly assigned to five dietary treatment groups (4 pigs per pen; 4 pens per treatment), including the basal diet (Control), basal diet supplemented with 1 (L), 5 (M), 20 (H) kg/ton of CMSSE, and antibiotics (PC) (100 ppm Tiamulin+220 ppm Oxytetracycline). The basal diet was formulated using a mixture of corn and soybean meal to meet nutrient requirements established by National Research Council standards (NRC, 1998).

The field trial was designed as a randomized complete block experiment, with four pigs per pen and four replications, spanning a duration of 8 weeks. During the experimental period, the pigs were provided with *ad libitum* access to both feed and water. Monthly data on body weight and feed intake were recorded to calculate the ADG and feed conversion rate (FCR). All pigs were immunized with a vaccine of swine mycoplasma *hyopneumoniae* (SEP, Ingelvac MycoFLEX®, Boehringer Ingelheim) in week 3, classical swine fever virus (CSFV,

Animal Health Research Institute, Council of Agriculture, Executive Yuan) in week 4 and *Actinobacillus pleuropneumoniae* (Porcilis APP, MSD) in week 6 and 9. Blood samples were collected from two randomly selected pigs within each replicate on days 0, 28, and 56 to assess the antioxidant activity and serum immune antibody concentrations.

Detection of pig immunoglobulin

The concentrations of IgA, IgG, and IgM were analyzed using the pig ELISA kits (#E100-102, #E100-104, and #E101-117, Bethyl Laboratories, Inc.). All procedures were performed in accordance with the instructions of the manufacturer.

Antioxidant analysis

Determination of plasma TEAC concentration followed established protocols [59, 60]. Briefly, ABTS and potassium persulfate oxidation reactions generated water-soluble ABTS+free radicals. These radicals reacted with antioxidants in the sample, inducing a change in the absorption spectrum. A trolox-based reference curve (0–500 µg/mL) at OD₄₀₅ facilitated the calculation of pig plasma antioxidant capacity. The TEAC concentration was determined using a regression equation. GPx activity in whole blood was measured with a commercial kit (#RS505, Randox Lab., London, UK) by monitoring the change at OD₃₄₀ over 0 to 3 min. Red blood cell SOD activity and heme concentration were analyzed using a kit (#10009055-Cayman Chemical, Cayman, Ann Arbor, MI, USA). Hemoglobin (Hb) was determined with a kit (#HG1539, Randox Lab., London, UK). Undiluted red blood cell samples were utilized, and purified bovine hemoglobin (#H2625-25G, Sigma, St. Louis, USA) served as the standard (0–0.08 g/mL) at OD₅₄₀. The sample concentration was calculated using a regression curve. For accuracy, SOD enzyme concentration was normalized with heme concentration, expressing SOD unit as U/g Hb.

Statistical analysis

The data were analyzed using an unpaired t-test or analysis of variance (ANOVA) in Prism software (GraphPad, San Diego, CA, United States), along with Duncan's multiple comparison test. $P < 0.05$ was considered statistically significant.

Abbreviations

AGPs	Antibiotic Growth Promoters
CMSSF	Cordyceps Militaris Solid-State Fermentation
CME	Cordyceps Militaris Ethanol Extract
DPPH	2, 2-Diphenyl-1-Picrylhydrazyl
ABTS	2,2'-Azinobis-(3-ethylbenzothiazoline-6-sulfonic acid)
IPEC-J2	Intestinal Porcine Enterocyte Cell Line
TNF	Tumor Necrosis Factor
IL	Interleukin

LPS	Lipopolysaccharide
SOD	Superoxide Dismutase
IgM	Immunoglobulin M
IgA	Immunoglobulin A
IgG	Immunoglobulin G
CSFV	Classical Swine Fever Virus
ROS	Reactive Oxygen Species
iNOS	Inducible Nitric Oxide Synthase
COX-2	Cyclooxygenase-2
CMF	C. Militaris Fermentation
HPLC	High-Performance Liquid Chromatography
TEAC	Trolox Equivalent Antioxidant Capacity
GPx	Glutathione Peroxidase
GC	Growth Control
NC	Negative Control
DMEM	Dulbecco's Modified Eagle's Medium/high glucose
PSA	Penicillin / Streptomycin / Amphotericin B
MTT	3-(4,5-dimethylthiazol-2-yl)-2,5 diphenyl
ADG	Average Daily Gain
ADFI	Average Daily Feed Intake
FCR	Feed Conversion Rate
CMS	Cordyceps militaris Spent Mushroom Substrate
MIC	Minimum Inhibitory Concentration

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Author contributions

CCP and CHC conceived and designed the experiments. CCP, YCC, and CCC performed the experiments and analyzed the data. JWL, CNL, MTC, and HWC contributed essential reagents and comments. CCP drafted the manuscript. JWL and CHC edited the manuscript. All authors read and approved the final manuscript.

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Data availability

The data and materials from this study can be obtained from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

All animal owners provided their informed consent for the participation of their animals in the experiments. The study was conducted in accordance with ARRIVE guidelines and approved by the institutional animal care and use committee (IACUC) of Agricultural Technology Research Institute (IACUC Approval No. 109021). All methods were carried out in strict accordance with relevant guidelines/regulations of the Ethics Committee on the Use and Care of Animals at Agricultural Technology Research Institute and National Pingtung University.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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